

A relativistic completion for the dark-energy–anchored acceleration scale $a_0 = \kappa c \sqrt{G\rho_\Lambda}$: aether–scalar–tensor theory with an offset-DBI khronon and an a_0 -resonant environment response

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We assemble and stress-test a relativistic field theory carrying the acceleration scale $a_0 = \kappa c \sqrt{G\rho_\Lambda} = 9.36 \times 10^{-11} \text{ ms}^{-2}$, where ρ_Λ is the measured dark-energy density and κ is a pure number that we treat as *measured*, not derived: a distance-free estimator built on the SPARC sample gives $\kappa = 0.551 \pm 0.043$ (± 0.063 if the H_0 tension is carried as a systematic), consistent with $\frac{1}{2}$ but not singling it out. The host is aether–scalar–tensor (AeST) theory [Skordis & Złośnik, PRL **127**, 161302 (2021)], the only relativistic MOND-class theory known to reproduce the CMB; the content of this work is the free function, which factorises into three terms with distinct jobs. (i) A MOND kernel normalised by a_0 , convex and bijective, with deep-MOND and Newtonian limits exact and a Newtonian residual of order $e^{-\sqrt{g/a_0}}$ ($\sim 10^{-3457}$ at Earth’s orbit). Because AeST has no gravitational slip ($\Phi = \Psi$, $\gamma_{\text{PPN}} = 1$), dynamical and lensing masses agree identically, converting the 21.2σ lensing exclusion that kills modified-inertia realisations into a 0.6σ pass. (ii) An offset Dirac–Born–Infeld (DBI) function of the khronon’s temporal invariant, whose minimum supplies $w = -1$ exactly (dark energy), whose quadratic deviations supply a pressureless cosmological component, and whose boundedness we show is *necessary*: for any power-law $K \sim u^n$ the early-time equation of state limits to $w = 1/(n - 1)$, so only a bounded K reaches dust. This reverses the sign of the published $455\times$ tension between cosmological and galactic requirements on the Helmholtz mass [Blanchet & Skordis, JCAP 11 (2024) 040], turning cosmology into a lower bound aligned with MOND ($226\times$ margin). A Boltzmann-code (CLASS) confrontation leaves TT unchanged to 0.069% and $P(k = 0.2 \text{ h/Mpc})$ to 1.7%. (iii) An environment response $AB(Y/a_0^2)(Q - Q_0)^2$ with $B(y) = y/(1 + y)^2$: a position-dependent Helmholtz mass peaked at the theory’s own a_0 , motivated by the observation that cluster interiors are the unique deep-potential environments sitting at the MOND transition (R_{500} at $0.33\text{--}0.58 a_0$). The term passes a five-environment matrix (clusters by calibration; galaxy interiors at 3.4×10^{-4} of the baryonic mass; galaxy outskirts at 1 Mpc at 1.2% against a 5.9% weak-lensing bound; the linear cosmos at 0.6% of mean matter; the solar system at 10^{-19}), vanishes identically on FRW, and enters linear cosmology at second order. A full perturbation analysis including all aether mixings yields: a no-ghost *theorem* (the term’s kinetic contribution is rank-one positive semi-definite, so by Weyl monotonicity it cannot lower any eigenvalue of the healthy AeST kinetic matrix); $c_T = 1$ exactly; a derived quasi-static closure ($\delta A^0 = -\Phi A^0$); and two-sided amplitude bounds that leave the mechanism falsifiable from both directions. The theory uses five dark-sector parameters against Λ CDM’s two; what is bought is that dark energy, the cosmological dark-matter phenomenology, and MOND arise from one function. We list eight explicit non-claims, led by “ $\kappa = \frac{1}{2}$ is not derived” and including the sharpest open problem: whether the dust component stays out of galaxy interiors, as the data demand, awaits a nonlinear relaxation calculation never performed for this theory class. Six falsifiable predictions follow, led by a hash-stamped, pre-registered Gaia DR4 wide-binary target. Every quantitative claim is reproducible from named, committed scripts that exit non-zero on failure.

I. INTRODUCTION

Two facts anchor this paper. First, galactic dynamics exhibits a characteristic acceleration: below $g \sim 10^{-10} \text{ ms}^{-2}$ rotation curves flatten, and the observed centripetal acceleration follows the baryonic prediction through a universal interpolation — the radial acceleration relation (RAR) [6, 7] — with remarkably small scatter. Milgrom identified the regularity in 1983 and codified it as MOND [1]. Second, the measured value of that scale is numerically close to accelerations constructible from the cosmological constant, as Milgrom also observed. The present program takes the second

fact literally and asks what follows if

$$a_0 = \kappa c \sqrt{G\rho_\Lambda} = \frac{cH_\Lambda}{Z} = c^2 \sqrt{\frac{\Lambda}{32\pi}} = 9.3619 \times 10^{-11} \text{ ms}^{-2}, \quad (1)$$

with $\kappa = \frac{1}{2}$, $Z = 2\sqrt{8\pi/3}$, and H_Λ the asymptotic de Sitter expansion rate, is a statement about gravity rather than a coincidence. The three forms in Eq. (1) are algebraically identical; κ and Z are conventions on one number.

A one-line relation is not a theory. It cannot bend light, form structure, or be checked for internal consistency. This paper assembles the relativistic field theory that carries Eq. (1) — “the completion” — and reports the outcome of an extended internal adversarial program:

uniqueness and no-go theorems, environment matrices, stability analyses, a Boltzmann-code confrontation, and the public withdrawal of three failed cluster mechanisms that preceded the surviving one. Throughout, results that could have killed the theory are given the same prominence as results that support it, and every quantitative claim is backed by a named script in the public repository [18] that recomputes it from scratch and exits non-zero on failure (Appendix A).

Three disclosures frame everything that follows. (1) The host theory, aether–scalar–tensor (AeST), is Skordis and Złośnik’s [5]; our content is the free function and the theorems about it. (2) An earlier arm of this program realised Eq. (1) through modified inertia; it is *excluded* at 21.2σ by the equality of dynamical and lensing masses and was withdrawn. The present modified-gravity realisation is the surviving arm. (3) κ is measured, not derived; Sec. II quantifies exactly how well, and Sec. IX lists what is not claimed.

II. THE SCALE: FORM, INTERPOLATION, COEFFICIENT

A. The form is forced

If a_0 is built from (G, c, ρ) alone, dimensional analysis admits exactly one combination, $a_0 = \xi c \sqrt{G\rho}$: the exponent matrix is nonsingular (determinant 2), so the form is unique, and the theorem has content because admitting $\hbar$ destroys the uniqueness. The choice $\rho = \rho_\Lambda$ is the physical hypothesis; the form then follows.

B. The interpolation is derivable — and its coefficient is excluded

A de Sitter–Unruh construction sharpens the stakes. Using the Deser–Levin temperature of an accelerated observer in de Sitter space [8] and Milgrom’s 1999 vacuum-heuristic [2], the MOND interpolation is *fully derived*, with no fitted element: $\nu(y) = \sqrt{1 + 1/y}$, which is precisely Milgrom’s Eq. (9) of Ref. [2]. The same construction, however, is rigid at every step — the power is forced by the two MOND limits, the baseline by the ambient Gibbons–Hawking temperature, the normalisation by the Newtonian limit — and it forces $a_0 = 2cH_\Lambda$, which the SPARC data exclude at 15.6σ . The construction that derives the *function* cannot supply the *coefficient*; the heuristic must never be cited as support for $\kappa = \frac{1}{2}$. Hence the coefficient must be measured.

C. The coefficient, measured

Two routes. The baryonic Tully–Fisher intercept on SPARC gives $\hat{\kappa} = 0.465 \pm 0.076$, with a hard 9.47% mass-budget floor (stellar and gas calibrations trade against

each other at fixed $f_\star + f_{\text{gas}} = 1$). A sharper, distance-free estimator exploits the fact that $g_{\text{bar}} \propto F/\theta^2$ (flux over angular size squared) carries no distance dependence while $g_{\text{obs}} \propto 1/D$ does; built on the g_{bar} axis of the RAR it gives

$$\kappa = 0.551 \pm 0.043 \quad (\pm 0.063 \text{ under the } H_0 \text{ tension}), \quad (2)$$

since $\kappa \propto a_0/H_0$ at full strength. Four simple candidates ($\frac{1}{2}, 1/\sqrt{3}, \sqrt{3/8}, 0.40$) sit inside 2σ : the data are consistent with $\frac{1}{2}$ and do not single it out. The binding systematics are the absolute 21-cm flux scale and CO-dark molecular gas, not stellar populations; an irreducible $\sim 3.9\%$ floor on a_0 caps the $\frac{1}{2}$ -vs- $1/\sqrt{3}$ discrimination near 4σ for any SPARC-class analysis.

III. THE ACTION

We adopt the AeST action of Ref. [5]: gravity is Einstein–Hilbert; matter couples to the metric only; the dark sector is a unit-timelike vector A^μ ($A^\mu A_\mu = -1$) and a scalar φ entering through the two invariants

$$Q \equiv A^\mu \nabla_\mu \varphi, \quad Y \equiv (g^{\mu\nu} + A^\mu A^\nu) \nabla_\mu \varphi \nabla_\nu \varphi. \quad (3)$$

The content of this work is the free function, which factorises:

$$\mathcal{F}(Y, Q) = \frac{a_0^2}{8\pi G} \mathcal{F}_Y\left(\frac{Y}{a_0^2}\right) + K(Q) + A B\left(\frac{Y}{a_0^2}\right) (Q - Q_0)^2 \quad (4)$$

with $B(y) = y/(1+y)^2$. The three terms are treated in Secs. IV, VI, and VII respectively.

The Y-sector. $\mathcal{F}_Y$ is fixed by the kernel $\nu(y) = 1/(1 - e^{-\sqrt{y}})$, $y = g_{\text{bar}}/a_0$, equivalently the closed parametric pair

$$\mu(u) = 1 - e^{-u}, \quad x(u) = \frac{u^2}{\mu(u)}, \quad u = \sqrt{y}, \quad x = \frac{g_{\text{obs}}}{a_0}, \quad (5)$$

normalised by Eq. (1).

The Q-sector. An offset-DBI function,

$$K(Q) = -M^4 + \mu^2 \Lambda_D^2 \left[1 - \sqrt{1 - \frac{u^2}{\Lambda_D^2}} \right], \quad u \equiv Q - Q_0, \quad |u| < \Lambda_D, \quad (6)$$

with $M^4 = \rho_\Lambda$ the vacuum scale, μ the Helmholtz mass, Λ_D the field-space cap, and the conserved shift charge I_0 setting the dust abundance.

The environment response. The third term is a position-dependent Helmholtz mass $\mu_{\text{eff}}^2 = A B(g^2/a_0^2)$ peaked at the theory’s own a_0 ; the peak location costs nothing (Y/a_0^2 is already the Y-sector’s normalisation), and the single new calibrated amplitude is $A \approx 1.7 \text{ Mpc}^{-2}$.

IV. GALACTIC PHENOMENOLOGY

The kernel (5) embeds exactly: the deep-MOND limit $\mu \rightarrow x$ and the Newtonian limit $\mu \rightarrow 1$ are exact, $x(u)$ is a bijection ($h' = e^{-u}(1+u) > 0$), and the resulting free function is convex, from which the baryonic Tully–Fisher relation follows exactly. The Newtonian residual is $e^{-\sqrt{y}}$: at Earth’s orbit, 3.6×10^{-3457} — planetary ephemerides are untouched at any conceivable precision. On the RAR, the framework’s own interpolation with the anchored a_0 of Eq. (1) fits SPARC at 0.108 dex scatter (best-fit $\Upsilon_\star = 0.70$), against 0.122 dex for the conventional literature interpolation with fitted a_0 : anchoring the scale to an external cosmological measurement removes a parameter without degrading the fit.

The environment response also operates inside galaxies and must not spoil this: it contributes 3.4×10^{-4} of the baryonic mass in galaxy interiors, shifting the RAR by 4×10^{-4} dex — two orders below the observed scatter (Sec. VII). A separate internal requirement — that the Q-sector dust of Sec. VI not accumulate in galaxy interiors — is open, and is stated as such in Sec. IX, non-claim 2.

V. LENSING

AeST has no gravitational slip: the weak-field potentials obey $\Phi = \Psi$, so $\gamma_{\text{PPN}} = 1$ and $M_{\text{dyn}}/M_{\text{lens}} = 1$ identically, for any source. This is the structural fact that separates the modified-gravity realisation of Eq. (1) from the modified-inertia realisation: the latter predicts $M_{\text{dyn}}/M_{\text{lens}} = 1/f_{\text{bar}} \simeq 6.4$ and is excluded at 21.2σ ; the former passes at 0.601σ . The dark field genuinely sources curvature where it accumulates, so light cannot distinguish this theory from a dark-matter universe at the level of the deflection law.

VI. COSMOLOGY: THE OFFSET-DBI SECTOR

A. Three regimes of one function

At the condensate minimum $u = 0$: $\rho = -K = M^4$, $p = K = -M^4$, hence $w = -1$ *exactly* — dark energy is the resting value of the khronon’s energy function, proved rather than assumed. For $0 < |u| \ll \Lambda_D$: $K \rightarrow \mu^2 u^2/2$ and the excitation dilutes as a^{-3} with vanishing pressure — the cosmological dust. As $|u| \rightarrow \Lambda_D$: K is bounded and $K' \rightarrow \infty$, and the early-time equation of state returns to $w \rightarrow 0$.

B. Boundedness is necessary: a no-go for power laws

For any power-law $K \sim u^n$ the early-time equation of state limits to

$$w \rightarrow \frac{1}{n-1}, \quad (7)$$

so the quadratic gives a stiff $w \rightarrow 1$ phase, the quartic $w \rightarrow \frac{1}{3}$, the sextic $w \rightarrow \frac{1}{5}$: every polynomial fails big-bang nucleosynthesis and CMB constraints in the same calculable way. Only a bounded K reaches $w \rightarrow 0$. This theorem re-frames the published $455\times$ conflict between the cosmological and galactic requirements on μ^{-1} [10], derived there for the quadratic form: with the DBI form the early stiff phase is absent and the sign of the cosmological requirement *reverses* — cosmology becomes a lower bound on μ^{-1} , aligned with the MOND-side requirement, with a $226\times$ margin. The no-go is not evaded but dissolved: its premise (the quadratic) is the one choice the theorem forbids. The DBI sector is ghost-free ($K'' = \mu^2(1-s^2)^{-3/2} > 0$ on the whole range) and subluminal ($c_s^2 \leq 0.385 \Lambda_D$). The cost is disclosed: an EFT boundary at saturation ($K''/\mu^2 \sim 3 \times 10^{11}$ at the cap), where the description is effective only.

C. Boltzmann-code confrontation

A real CLASS run with the DBI sector mapped to a generalised-dark-matter fluid gives, at $\Lambda_D \leq 10^{-2}$ (natural units of Ref. [5]): $c_s^2(z_{\text{rec}}) = 2.9 \times 10^{-8}$, TT spectrum unchanged to 0.069%, and $P(k = 0.2 h/\text{Mpc})$ to 1.71% — indistinguishable from CDM on the primary CMB. Two caveats are part of the result. First, the pass depends on the sound-speed bump being transient (it peaks at $z \approx 189$); a patched fluid carrying the full $c_s^2(a)$ history through structure growth is required and owed. Second, the stability analysis of Sec. VII C subsequently forced $\Lambda_D \leq 8.4 \times 10^{-7}$, *excluding the fiducial value used in the CLASS run*; since the run’s agreement only improves with smaller Λ_D (the sound speed falls), the pass survives, but the window $1.9 \times 10^{-10} \ll \Lambda_D \leq 8.4 \times 10^{-7}$ (3.7 decades) is now health-bounded from above.

D. What the CMB requires, exactly

The CMB requires a component that gravitates, is pressureless, and dilutes as a^{-3} , at the full Ω_{dm} : deleting it moves the third-to-first acoustic peak ratio by 54% and no refit absorbs it ($\Delta\chi^2 > 400$). It does not require a particle: by the generalised-dark-matter degeneracy [14], a fluid excitation and a particle species are indistinguishable at 0σ in the linear CMB. This theory supplies the component as the khronon’s shift-charge excitation. The defensible statement — and the only one

this paper makes — is: *no dark-matter particle; the cosmological pressureless job is done by the same field that supplies Λ and MOND*. The flat claim “there is no dark matter” is not available to this theory or to any theory that matches the CMB. Nor, at present, is “none in galaxies”: the galaxy data are fit by baryons plus the kernel alone, so the theory *needs* its dust component to stay out of galaxy interiors, and whether it does is an open nonlinear problem (Sec. IX, non-claim 2).

VII. CLUSTERS: THE a_0 -RESONANT ENVIRONMENT RESPONSE

A. Why the obvious routes fail

MOND under-predicts cluster masses; any completion must supply the residual without disturbing galaxies that share the relevant scales. Three mechanisms were built and killed within this program, and the kills shape the survivor:

(a) *The R^2 lever* (residual density tracking the potential, ratio fixed by system size) requires $\mu^{-1} = 3.13$ Mpc, violating both galaxy-outskirt weak-lensing bounds of Ref. [13]. Withdrawn.

(b) *The primordial shift-charge route* (clusters khronon-rich because their Lagrangian patches were over-dense) fails three independent ways, the third decisive: (i) cluster R_{500} and galaxy-outskirt lensing sample wavenumbers only $1.4\times$ apart ($k \simeq 4.5$ vs 6.3 Mpc^{-1}) with disjoint abundance requirements; (ii) for correlated Gaussian initial conditions $\mathbb{E}[\delta_{\text{kh}}|\delta_m]$ is linear in δ_m , so no such statistic can select environments; (iii) *smooth accretion*: any suppression mechanism removes fluctuations, not the mean, and a collapsing region captures all cold content in its basin (the khronon obeys the identical shell equation by the equivalence principle), so $\xi(\text{halo}) \rightarrow 1$ for *any* cold transfer function. Withdrawn, and the prior version of this paper’s cluster section with it.

(c) *The potential response* $\rho_c \propto \Phi^n$: cosmological potentials are scale-free ($\Phi_{\text{rms}} \sim 10^{-5}$ everywhere, only $2.2\times$ below cluster centres), so any n steep enough to select clusters floods the linear cosmos at $\sim 20\times$ the mean matter density. Closed.

The lesson: neither scale nor depth alone selects clusters. What does is the pair (depth, $g \sim a_0$): cluster interiors at R_{500} sit at $0.33\text{--}0.58 a_0$ — *clusters are the cosmic objects that live at the MOND transition*.

B. The term, and the environment matrix

The third term of Eq. (4) implements exactly that selection: a Helmholtz mass $\mu_{\text{eff}}^2 = AB(g^2/a_0^2)$ with $B(y) = y/(1+y)^2$ peaked at $g = a_0$, falling on both sides. Calibrating $A \approx 1.7 \text{ Mpc}^{-2}$ on the cluster residual ($\mu_{\text{eff}}^2 = 0.23 \text{ Mpc}^{-2}$), the term then passes the four non-calibration environments: galaxy interiors, 3.4×10^{-4} of

M_b (B near peak but the potential shallow); galaxy outskirts at 1 Mpc, 1.2% of the baryonic mass against the 5.9% strict bound from Ref. [13] stacking; the linear cosmos, 0.6% of mean matter; the solar system, 10^{-19} . On FRW, $Y = 0$ and the term vanishes identically; it enters perturbations at second order, so the linear CMB and $P(k)$ of Sec. VI are untouched by construction. It evades kill (b) because a response has no charge to advect, and kill (c) because it does not couple to depth alone. Against interest: cluster modelling in Ref. [13]’s own framework prefers amplitudes $4\text{--}34\times$ our chain calibration; the health bounds below cut into exactly that band, making the tension a two-sided test rather than a tuning freedom.

C. Stability, and the full perturbation matrix

No ghosts, as a theorem. The term’s contribution to the kinetic matrix is $2AB(y) \geq 0$ in the $\chi\chi$ entry alone — a rank-one positive semi-definite addition. By Weyl’s monotonicity theorem, adding a PSD matrix cannot lower any eigenvalue; AeST’s kinetic matrix is healthy in its stable region; therefore the completed theory is ghost-free wherever AeST is, *for all field values and all environments*. Verified in closed form on the 2×2 subsystem and numerically on random healthy 4×4 matrices. There is no Ostrogradsky sector (no higher derivatives are introduced).

Tensor speed. The new terms are algebraic in the metric perturbation: $c_T = 1$ exactly, and GW170817 [15] is satisfied by construction, not by tuning.

Quasi-static closure. The unit-norm constraint gives $\delta A^0 = -\Phi A^0$, hence $\delta Q = \dot{\chi} - Q_0\Phi$; the full matrix regenerates the quasi-static phenomenology assumed in the environment matrix. The gradient block including all scalar–aether mixings is, exactly,

$$\Delta G = 2\lambda \begin{pmatrix} q & 2y^{3/2}B'' \\ 2y^{3/2}B'' & yq \end{pmatrix}, \quad q = \frac{3y^2 - 8y + 1}{(1+y)^4}, \quad (8)$$

with λ the dimensionless bump amplitude in the Y-sector’s gradient normalisation (exact assembly in the scripts). $q < 0$ precisely on $0.13 < y < 2.5$ — the cluster band — which is what turns stability into a cap on A .

The bounds. On FRW, gradient health forces $\Lambda_D \leq 8.4 \times 10^{-7}$, overruling this program’s own earlier fiducial 10^{-2} ; the Φ -mixing correction is Poisson-suppressed (3×10^{-4}) and does not reopen it. In halos, the isolated-term analysis caps $A \leq 2.7 A_{\text{fid}}$; integrating out the aether softens this to $A_{\text{max}} = (2.7\text{--}7.4) A_{\text{fid}}$, the residual range pinned by one coefficient of the quasi-static expansion of Ref. [5]’s supplemental material (a literature lookup). Either way, the demanding ($34\times$) end of the Ref. [13] band is excluded by $> 7\times$ and the $4\times$ edge is marginal: the amplitude is pinched between an observational floor and a stability ceiling. The fiducial calibration sits inside but near the edge ($1 + 2\lambda q = +0.63$

at the cluster minimum). Two flags remain open and are so labelled: a λ -suppressed tachyon-type vector mass in $y > 1$ interiors (watch item), and the known $k < \mu$ formal unboundedness of AeST [5], which at this theory's $\mu^{-1} = 4392$ Mpc migrates to the Hubble scale (within 15%) — relocated, not resolved.

Free signature. The response carries anisotropic stress of order 0.6 in clusters: a specific lensing-vs-hydrostatic mass discrepancy, testable on existing samples (Sec. X).

Status. The mechanism is a *candidate*: it has survived this program's environment matrix, health check, and full perturbation analysis, and no external scrutiny. A real cluster-model fit is owed.

VIII. PARAMETER COUNT, HONESTLY

Λ CDM's dark sector: Ω_c, Ω_Λ . This completion: M ($= \rho_\Lambda^{1/4}$), I_0 , μ , Λ_D , A — *five numbers against two*, with a_0 not independent: $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$, $m_{\text{cond}} = M^2/(\sqrt{2}M_{\text{Pl}})$, holds to 0.076% — but see non-claim (1) below for why this is a relabelling rather than a derivation. What the extra parameters buy is structural: Λ , the cosmological dust, and MOND from one function instead of three unrelated sectors, plus kill-tests Λ CDM does not offer.

IX. WHAT IS NOT CLAIMED

1. $\kappa = \frac{1}{2}$ **is not derived.** $a_0 = m_{\text{cond}}/(4\sqrt{\pi})$ is algebraically identical to $a_0 = \frac{1}{2}c\sqrt{G\rho_\Lambda}$ and to $Z = 2\sqrt{8\pi/3}$; the residue is convention-dependent (reduced vs non-reduced Planck mass), and a derived number cannot depend on bookkeeping. The best current framing is a graviton-bath calculation in which the de Sitter horizon entropy cancels the Planck suppression exactly ($S_{\text{ds}} GH^2 = \pi$ identically), leaving $\kappa^2 = 8\pi\epsilon_{\text{tot}}$ with ϵ_{tot} a pure number from horizon-mode counting — well-posed and open. Five defensible normalisations span $\kappa = 0.013$ –2.047; the one landing exactly on $\frac{1}{2}$ is structurally invalid ($h_{\mu\nu}u^\mu u^\nu$ vanishes in TT gauge for a static worldline) and is not cited.
2. **“No dark matter” is not claimed — and “none in galaxies” is an open problem, not a result.** The pressureless component exists at full Ω_{dm} (Sec. VID); what does not exist is a particle. Whether the dust stays out of galaxy interiors is genuinely open: the smooth-accretion theorem that killed cluster route (b) implies it falls into every collapsing basin, galaxies included. The favorable branch — relaxation to the Helmholtz-preferred, centrally-evacuated static profile (ρ_c tracking the potential, $\xi \propto R^2$, galaxy-interior cost $\sim 6 \times 10^{-5}$ dex) — and the fatal branch — CDM-like concentration, overshooting the RAR by the banked 2.06–

4.42× — are both computable; the deciding quantity is the condensate's relaxation time toward the static profile within a Hubble time, a nonlinear calculation never performed for AeST by any group. It is this program's next scheduled computation. Until then, galaxy-interior cleanliness is an assumption the theory needs, not a result it owns.

3. **The cluster mechanism is a candidate,** pinched and falsifiable, not a result.
4. **The post-recombination growth history is partially verified.** The acoustic-peak pass is real; a patched $c_s^2(a)$ fluid through structure growth is owed, as is a Lyman- α confrontation of the dust component's small-scale initial conditions.
5. **Cassini Q_2 is inherited** (3–15 σ class tension) and not relieved here.
6. **PPN α_1, α_2** for this free function are not yet computed against lunar-laser-ranging and binary-pulsar bounds.
7. **The wide-binary target is the point-field asymptote;** the full nonlinear AQUAL-EFE solve is owed before Gaia DR4.
8. **This is not a theory of everything;** nothing here touches the Standard Model. The 2026-06-23 retraction of earlier overclaims stands.

X. FALSIFIABLE PREDICTIONS

1. *Wide binaries (Gaia DR4).* Registered, hash-stamped target $\gamma_v = \sqrt{\nu(y_{\text{ext},N})} = 1.2139$ (canonical footing; 1.2592 alternative), against the withdrawn modified-inertia arm's 1.1582 — separated at 2.68 σ by the frozen pipeline's own error budget. The pre-registration and nine hash-stamped amendments are committed [18]; DR4-as-frozen measures a_0 to 12.6%.
2. *Directional external-field asymmetry.* AQUAL-class prediction 1–4%, signed; pure modified inertia predicted exactly zero. A first firing on existing data gives $\hat{A} = +2.95$, $p = 0.029$, with the predicted sign — suggestive, not decisive.
3. *Cluster anisotropic stress.* Order-0.6 lensing-vs-hydrostatic mass discrepancy with a specific profile — the bump's free signature.
4. *Cluster-to-cluster scatter* in the residual, $\sigma(\xi)/\xi \approx 0.02$ –0.60, distinguishing the environment response from mechanisms predicting zero scatter, on existing data.
5. $a_0(z)$. If $a_0 \propto \sqrt{\rho_{\text{de}}(z)}$, then $a_0(z)/a_0(0) = (1+z)^{\frac{3}{2}(1+w_0+w_a)}e^{-\frac{3}{2}w_a z/(1+z)}$ — a bump-then-decline

under DESI-class (w_0, w_a), distinct from constancy and from $cH(z)$ scaling.

6. *No particle, ever.* Any confirmed direct detection of a dark-matter particle falsifies the theory outright.

XI. CONCLUSION

The relation $a_0 = \kappa c \sqrt{G\rho_\Lambda}$ now has a relativistic carrier: AeST with the factorised free function of Eq. (4). Within it, lensing, the primary CMB, the RAR, the BTFR, and the solar system are quantitatively passed; dark energy is the condensate minimum rather than an input; the published quadratic-sector no-go is dissolved by a boundedness theorem; and the long-standing cluster problem has, for the first time in this program, a candidate mechanism that survived its own execution squad — resonant, falsifiably, at the theory’s own acceleration scale. The theory is more expensive than Λ CDM in parameters and more constrained in structure; its distinctive wager is that one measured coincidence is a law. The wager is now fully specified, stability-certified, and scheduled for execution or vindication by data that exist or are imminent.

Appendix A: Reproducibility

Every quantitative claim above is recomputed by a committed script in Ref. [18]

(`real_research/reviews/`), each of which exits non-zero on failure and carries negative controls. Key mappings: lensing and the AeST embedding, `mi_relativistic_completion_aest_2026.py`; the DBI sector, its theorem and health, `mi_dbi_khronon_2026.py`; the CLASS run, `mi_dbi_cmb_class_run_2026.py`; the CMB dust requirement, `mi_cmb_no_dust_existence_2026.py`; the condensate- a_0 relation, `mi_condensate_vacuum_energy_a0_2026.py`; the Deser-Levin derivation and its 15.6σ scissors, `mi_deser_levin_interpolation_2026.py`; the coefficient measurement, `mi_distance_free_gbar_estimator_sparc_2026.py` and `mi_kappa_error_budget_unlock_2026.py`; the cluster-route kills, `mi_ic_route_1mpc_confrontation_2026.py`; the environment response, its health, and the full perturbation matrix, `mi_a0_bump_response_2026.py`, `mi_a0_bump_health_2026.py`, `mi_aest_full_matrix_bump_2026.py`; the graviton-bath framing of κ , `mi_graviton_bath_ctp_2026.py`; kernel embedding and BTFR, `mi_route_a_field_theory_2026.py`.

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